

Fractions & Decimals

Number World Study Notes

Numerator, denominator, simplifying, and converting between fractions, decimals, and percents

Fractions and decimals are two different ways of writing numbers that are not whole. Learning to move confidently between fractions, decimals, and percents makes it much easier to compare quantities, solve real-world problems, and work with proportions.

Key Definitions

Fraction

A number written as a/b , where 'a' is the numerator (parts we have) and 'b' is the denominator (total equal parts), b not equal to 0.

Proper, Improper & Mixed

Proper fraction: numerator < denominator (e.g., $3/4$). Improper fraction: numerator $\geq$ denominator (e.g., $7/4$). Mixed number: a whole number plus a proper fraction (e.g., $1 \frac{3}{4}$).

Equivalent Fractions

Fractions that represent the same value even though they look different, formed by multiplying or dividing the numerator and denominator by the same non-zero number.

Simplifying (Reducing)

Dividing the numerator and denominator by their Highest Common Factor (HCF) to write a fraction in its lowest terms.

Conversion & Comparison Rules

- Fraction to decimal: divide the numerator by the denominator (a / b).
- Decimal to fraction: write the decimal digits over the matching place-value denominator (10, 100, 1000, ...), then simplify.
- Fraction or decimal to percent: multiply by 100 and add the % sign.
- Percent to fraction or decimal: divide by 100.
- Comparing fractions: convert to a common denominator (LCD), then compare numerators directly.
- A fraction is in simplest form when the HCF of its numerator and denominator is 1.

Worked Examples

Example 1: Simplify $18/24$.

Find the HCF of 18 and 24, which is 6.

Divide both by 6: $18/6 = 3$, $24/6 = 4$.

Answer: $3/4$

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Example 2: Convert $\frac{3}{8}$ to a decimal.

Divide numerator by denominator: $3 \div 8 = 0.375$.

Answer: 0.375

Example 3: Convert 0.6 to a fraction in simplest form.

$0.6 = \frac{6}{10}$ (one decimal place $\rightarrow$ denominator 10).

Simplify by dividing both by their HCF, 2: $\frac{6}{10} = \frac{3}{5}$.

Answer: $\frac{3}{5}$

Example 4: Compare $\frac{2}{3}$ and $\frac{3}{5}$ — which is larger?

Find the LCD of 3 and 5, which is 15.

Convert: $\frac{2}{3} = \frac{10}{15}$ and $\frac{3}{5} = \frac{9}{15}$.

$\frac{10}{15} > \frac{9}{15}$.

Answer: $\frac{2}{3}$ is larger

Practice Questions

- 1 Simplify $\frac{45}{60}$ to its lowest terms.
- 2 Convert $\frac{7}{20}$ to a decimal and then to a percent.
- 3 Convert 1.25 to a mixed number.
- 4 Which is larger: $\frac{5}{8}$ or 0.6?
- 5 Add: $\frac{1}{4} + \frac{1}{6}$ (give the answer as a simplified fraction).

Answer Key

- 1 $\frac{3}{4}$ (HCF of 45 and 60 is 15).
- 2 0.35, which is 35%.
- 3 $1\frac{1}{4}$ (since $1.25 = 1 + \frac{25}{100} = 1 + \frac{1}{4}$).
- 4 $\frac{5}{8}$, since $\frac{5}{8} = 0.625$ which is greater than 0.6.
- 5 $\frac{5}{12}$ (LCD is 12: $\frac{3}{12} + \frac{2}{12} = \frac{5}{12}$).